

45 Days DSA & Logical Thinking Roadmap (Pattern Wise)

This roadmap reorganizes the original 45-day PHP logical coding checklist into DSA patterns. Focus on pattern recognition instead of memorizing solutions.

Pattern 1 — Basic Array Traversal & Tracking

Goal: Loop mastery and traversal thinking.

#	Question
1	Print all even numbers from 1 to N
2	Print all numbers divisible by 3 and 5
3	Implement Linear Search manually
4	Find second largest element
5	Find second largest with duplicates
6	Count even and odd elements in array
7	Find sum of array elements
8	Check if number exists in array
9	Find maximum element
10	Find minimum element

Pattern 2 — HashMap / Associative Array / Frequency

Goal: Key-value and frequency counting logic.

#	Question
1	Count frequency of array elements
2	Count frequency of characters in string
3	Print unique elements from array
4	Remove duplicate elements from array
5	Find first repeating element
6	Find first non-repeating element

Pattern 3 — Reversal & Two Pointer

Goal: Reverse traversal and pointer thinking.

#	Question
1	Reverse array manually
2	Reverse string without strrev()
3	Check palindrome number
4	Check palindrome string
5	Swap two numbers without third variable

Pattern 4 — Number Logic & Math Thinking

Goal: Modulo/division and mathematical logic.

#	Question
1	Armstrong number
2	Sum of digits
3	Count digits in integer
4	Factorial without recursion
5	Prime number check
6	Print first N primes
7	Perfect square check without sqrt()
8	Fibonacci series
9	Check Fibonacci number
10	GCD of two numbers
11	LCM of two numbers

Pattern 5 — String Processing

Goal: Character traversal and manipulation.

#	Question
1	Count vowels and consonants
2	Check if two strings are anagrams
3	Count character frequency

Pattern 6 — Pattern Printing

Goal: Nested loop mastery.

#	Question
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1	Triangle star pattern
2	Inverted triangle pattern
3	Half pyramid using numbers
4	Multiplication table

Pattern 7 — Array Rearrangement & Rotation

Goal: Rearrangement and index manipulation.

#	Question
1	Merge two arrays manually
2	Rotate array right by K
3	Rotate array left by K
4	Move all zeros to end
5	Separate positive and negative numbers
6	Find intersection of two arrays

Bonus Interview Pattern

Goal: Real interview optimization thinking.

#	Question
1	Find all pairs whose sum equals target
2	Find missing number from 1 to N

Important: Strong developers recognize patterns instead of memorizing questions.